

Jiaxin (Jeremiah) Pan

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Portfolio · GitHub · LinkedIn

Research Profile

Physics undergraduate working on AI for Science, generative sampling, interpretable materials representations, and language-model reasoning. Research interests connect statistical mechanics with reproducible scientific computing and reliable machine-learning methods.

Education

2023–2027

B.Sc. in Physics (with Honours) — First Class Honours expected

The Chinese University of Hong Kong, Shenzhen · Shenzhen, China

GPA: 3.61/4.0. Four-year honours degree with a required research thesis. Admitted to Economics in 2023; completed Year 1 foundation courses before transferring to the B.Sc. Physics programme in 2024 to pursue my passion for sciences.

Exchange and Visiting Programmes

2026 (Spring)

Visiting Student

University of Auckland · Auckland, New Zealand

Nominated by CUHK-Shenzhen; completed advanced coursework in Physics, Computer Science, and AI, including COMPSCI 361 and COMPSCI 220.

2025

(Summer)

Summer Visiting Student

University of California, Berkeley Extension · Berkeley, CA, USA

Completed coursework in physics and neuroscience; developed cross-cultural academic communication and independent learning skills.

Ongoing Research

Ongoing

Data-Free Generative Sampling of the Two-Dimensional XY Model

Undergraduate Researcher, CUHK-Shenzhen

Supervised by Prof. Kai Zhou. Provisional portfolio title, not a finalized thesis title.

- Contributed to adapting Stochastic Path Sampling (SPS) to periodic XY angle variables. Training uses the XY energy and forward/backward path probabilities without pre-generated equilibrium configurations as training data. Generated paths are evaluated with path-space independence Metropolis–Hastings (MH) correction.
- Earlier exploration used supervised Flow Matching trained on Wolff-generated configurations. Wolff sampling is an independent validation reference for the current method.
- Completed a training-budget comparison across three seeds, 5,000/10,000/15,000 parameter updates, and 32/64/128 generation steps. Python/PyTorch GPU experiments run through Slurm.
- Evaluated energy, vortex counts, magnetization, spatial correlations, acceptance, effective sample size per sampling second, and consecutive rejection streaks.
- Estimated energy sampling efficiency increased in all nine seed/step settings from 5,000 to 10,000 updates; from 10,000 to 15,000, six improved and three declined. All nine settings at both later checkpoints passed predefined physical-statistics screening criteria; this supports, but does not prove, complete equilibration.
- High relative path weights can cause prolonged MH rejection streaks. Acceptance alone does not determine sampling efficiency. Analysis and report preparation are underway.

The limited 4×4 lattice study at one temperature near the expected Berezinskii–Kosterlitz–Thouless (BKT) transition does not establish thermodynamic critical behavior, a solution to critical slowing down, universal convergence, or an advantage over Wolff. SPS is an existing method.

Ongoing **Interpretable Crystal Representations for AI-Assisted Materials Design**

HSSRLM — AI for Science — Working paper

An initial AAAI submission plan was deferred while another manuscript took priority. Possible future venues include IJCAI and ACL, subject to fit and further development. A collaborator indicated that my current contributions could support co-first authorship; co-first, second, or other authorship positions depend on final contributions and author agreement. Submission plans and final author order remain unconfirmed. Framework: compact hierarchical descriptors form an interpretable interface between natural language and porous materials, including metal–organic frameworks (MOFs) and zeolites.

- Hierarchical Structure-Sequence Representation (HSSR) organizes composition, lattice parameters, symmetry, topology, building units, and selected pore-related properties into structured text. The repository implements T5 (Text-to-Text Transfer Transformer) infrastructure for bidirectional text/descriptor translation.
- Contributed to evidence-aware Crystallographic Information File (CIF)-to-descriptor preprocessing that separates known, inferred, and missing information, preserves provenance, and routes incomplete records for review.
- Contributed to multi-candidate screening with descriptor validation, node/linker/topology connectivity checks, explicit failure reporting, and descriptor-level duplicate tracking.
- Contributed to reproducible structure-equivalence audits on three publicly sourced MOF structures, examining equivalent representations and information lost by coarse descriptors. This is a limited study, not evidence of universal representation invariance.

Framework implementation and targeted reliability audits are complete; broader dataset validation and model evaluation remain in progress. HSSR omits complete atomic coordinates and a full bonding graph, so its outputs are descriptors rather than complete atomistic CIF structures. Verification establishes compatibility with implemented rules, not physical validity or synthesizability. CrysSpec (Crystal Specification) is a provisional working name. The framework also includes a 26-record metal-node registry that addresses ambiguous aliases and preserves source material for continued chemical validation.

Additional Research Experience

2026 (Spring) **Remote Research Assistant**

INSAIT · Bulgaria / Remote
Under Dr. Jinjin Gu.

- Analysed large language model reasoning and post-training, including Surgical Post-Training (SPOT), implicit regularization, minimal-edit data rectification, and catastrophic forgetting.
- Studied the trade-off between reasoning performance and foundational knowledge retention; prepared research syntheses and contributed to seminars.

2025 (Fall) **Guided Research on Van der Waals Integration and 2D Electronics**

CUHK-Shenzhen, Prof. Qi Qian's Lab

Participated in fabrication and characterization of MoS₂ transistors through van der Waals integration and dry transfer of 2D heterostructures. Studied metal–semiconductor contacts and polarity modulation; gained hands-on nanofabrication and electrical characterization experience, conducted literature review, and prepared an end-of-term report.

Earlier work **Guided Research Training in Generative AI and Diffusion Models**

CUHK-Shenzhen, Prof. Kai Zhou's Group

Earlier training in score-based generative modelling and physical dynamics.

- Explored connections between stochastic differential equations and diffusion models, including the denoising score-matching objective.
- Developed simulations of a one-dimensional double-well potential and visualized generative trajectories.
- Analysed learned score fields and local stability using Jacobian matrices; compared Langevin-sampled steady-state distributions with the Gibbs distribution.

Conference

31 Jan 2026 **CUHK-Shenzhen Undergraduate Research Conference (URC2026)**

Second Undergraduate Research Conference · Shenzhen, China

Presented “Self-Confidence in Math and STEM Major Choice at CUHK-Shenzhen” at my first research conference. The study examined the relationship between mathematical self-confidence and STEM major choice. [Research paper \(public copy\)](#); presentation certificate and on-site photographs on the portfolio.

Course Projects

- 2025 (Fall) **Group Research on Bose–Einstein Condensation**
CUHK-Shenzhen
Group leader. Coordinated literature study of “Observation of Bose-Einstein Condensation in a Dilute Atomic Vapor”; delegated tasks, co-authored LaTeX presentation slides, and delivered the class presentation.
- 2025 (Spring) **Group Research on Bell Test in Quantum Physics**
CUHK-Shenzhen
Group leader. Coordinated a study of experimental verification of Bell's theorem; delegated tasks, co-authored the final report, and delivered the presentation.
- 2025 (Spring) **Group Research on Computational Physics: Quantum Harmonic Oscillator**
CUHK-Shenzhen
Used finite-difference discretization and matrix eigenvalue methods for the one-dimensional time-independent Schrödinger equation. Compared numerical and analytical solutions, examined convergence with grid resolution and domain width, and synthesized team members' results into a final LaTeX report.
- 2024 (Fall) **Group Research on Semiconductor Optical Properties**
CUHK-Shenzhen
Group leader. Coordinated a study of semiconductor optical properties; conducted literature review, authored the final report, and delivered the class presentation.

Teaching and Work Experience

- Oct–Dec 2025 **Undergraduate Student Teaching Fellow (USTF)**
CUHK-Shenzhen School of Science and Engineering
Provided tutoring in the Student Help Room, a form of USTF service, for CSC1005 (Introduction to Computer Engineering), CSC1001 (Introduction to Computer Science), and PHY1001 (Mechanics), 23 October–12 December 2025. Help Room certificate available on the portfolio.
- 2025–2026 **Study Abroad Student Ambassador**
CUHK-Shenzhen Office of Academic Links · Australia and New Zealand Group
Served as a liaison between students and OAL for communication, safety reminders, and emergency handling; supported students' academic and living experiences, organized group activities, and contributed to outbound-study sharing sessions. Recognized by OAL for exemplary performance.
- 2026 (Spring) **Study Abroad Instagram Ambassador**
University of Auckland · Auckland, New Zealand
March–June 2026. Produced multimedia content for the official Study Abroad Instagram account; recognized by the Study Abroad Team as the most prolific ambassador with the highest total likes across posts. Recommendation letter available on the portfolio.
- 2023–2025 **Private Tutor**
Self-employed
Tutored CET-4, college entrance English, and international middle-school mathematics using the Big Ideas Math curriculum; tailored lesson plans to students' needs.
- 2023 (Summer) **Teaching Assistant**
New Oriental Education and Technology Group · Chongqing, China
Supported English instruction, classroom management, teaching-material preparation, and communication with parents.

Honours and Awards

- 2023–2026 **Full-Tuition Scholarship**
CUHK-Shenzhen
Philip Yu Hong Wong Anita Fung Yee Leung Full-tuition Scholarship. Awarded based on outstanding performance in the 2023 Gaokao.

2026 **Dean's List**
CUHK-Shenzhen School of Science and Engineering
Recognized for academic excellence in AY2025–2026. Certificate pending issuance.

2025 **Dean's List**
CUHK-Shenzhen School of Science and Engineering
Recognized for academic excellence in AY2024–2025.

2024 **Dean's List**
CUHK-Shenzhen School of Management and Economics
Recognized for academic excellence in AY2023–2024.

2024–2025 **Shaw Overcoming Adversity Award**
CUHK-Shenzhen Shaw College
Recognized for resilience and perseverance in overcoming personal and academic challenges.

Certification

2025 HCIA-AI: Huawei Certified ICT Associate — Artificial Intelligence. Issued by Huawei Technologies Co., Ltd.; valid through December 2028. Certificate No. 010102001810062372927192.

Skills

Scientific computing	Python, PyTorch, NumPy, SciPy, Matplotlib, CUDA, Git, LaTeX; GPU experiment workflows through Slurm. Coursework includes CSC1001 (A grade), CSC1006 (machine-learning concepts), and PHY2650 (team project awarded full marks plus an excellence bonus).
Machine learning	Diffusion models, Flow Matching, and Diffusion Transformers (DiT) for AI-for-Science applications.
Collaboration	Microsoft Word, Excel, PowerPoint, Teams, and Planner.
Content creation	Instagram, WeChat Official Accounts, and TikTok; photo/video editing with iMovie and TikTok Studio.

Languages

English	Full professional proficiency. TOEFL iBT: 100 (Reading 29, Listening 26, Speaking 23, Writing 22), October 2025.
Mandarin	Native proficiency.
Cantonese	Basic conversational ability.

Extracurricular Activities

Music and the arts	An unwavering love of the arts alongside my studies and research, expressed through choral singing and public performance.
2026 (Fall)	CUHK-SZ Chorus Choir Member Joined the university chorus in Fall 2026, continuing my commitment to choral music.
2026 (Spring)	Auckland University Student Choir (AUSC) Tenor · University of Auckland exchange Performed in two public concerts: the ANZAC Eve Commemorative Concert (24 April 2026) and <i>Oh, the Places We'll Go</i> (29 May 2026). Performance record ; six-video concert playlist .
2025 (Fall)	School of Music Choir, CUHK-SZ Choir Class Member Participated in the choir class and performed in the end-of-term concert, <i>Winter Carols</i> , on 12 December 2025.
2024–Present	CUHKSZ Economics Club Core Management; Head of Secretary Department Managed documentation, social media content, and event logistics.
2024–Present	CUHKSZ Physical Society Management Organized academic events and coordinated internal operations.
2023–2024	CUHKSZ Shaw College Student Helper Created WeChat content, provided multimedia support, and assisted in campus-event organization.
2023	CUHKSZ School of Humanities and Social Sciences Research Assistant Supported research documentation through field photography and video editing; assisted professors with data organization.